PROJECT PROPOSAL

Team 20

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Medical Insurance Cost Prediction

Objective

1. Analyze Medical Insurance Cost Prediction (of 100,000) patients, and classify said patients' healthcare insurance based on a multitude of factors:
    a. Demographic
    b. Socioeconomic Status
    c. Insurance plans
    d. Lifestyle

    From a low-high quality insurance standing

Rationale

Medical insurance costs are influenced by a wide range of demographic, lifestyle, clinical factors, making cost prediction a challenging and important problem. Accurately predicting medical insurance expenses can help improve risk assessment and support data-driven decision making in healthcare and insurance systems. Due to the complex relationships among these factors, machine learning methods are well suited for this task.

Model/Approach

In order to effectively classify our data, applying logistic regression to all relevant variables is a great place to start, also applying regularization on variables such as age, marital status, doctor visits, etc., and including interaction terms for said variables for more flexibility.

Timeline

Week 3(01/25): Project proposal complete and submitted

Week 4(02/01): Data set features and targets decided. EDA begins

Week 5(02/08): Relevant research concluded, EDA done

Week 6(02/15): Mid-Quarter Report done, early models completed

Week 7(02/22): All models finalized, begin writeup

Week 8(03/01): Project Done. Website created

Team Responsibilities

- Team Leader: Manage schedules and task assignments; organize and supervise the stable progress of the project, and facilitate communication between team members; ensure reproducibility and the final submission files**. (Mohsin)**
- Model Creator: Develop the basic model, and gradually maintain and advance higher-level models to achieve objectives; perform tuning and final model selection; maintain training and evaluation scripts. **(Ben, Xiang Mao)**
- EDA: Conduct exploratory analysis; generate visualizations; identify data quality issues, missing values, outliers, data leakage, and provide feature engineering recommendations. **(Lu Liang)**
- Report Writer: Write and edit reports, meeting the report requirements based on the rubric, ensuring the authenticity and objectivity of the report. Complete the work according to a planned schedule before the deadline, and review and revise with teammates to ensure accuracy. (**Nelson**)